RESEARCH ARTICLE

Agricultural Production Planning under Correlated Price Risk: Exact Reference Computation and Surrogate-Guided Continuous Refinement

Leonardo Hernán Talero-Sarmiento

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ABSTRACT
Agricultural production planning couples discrete crop calendars with continuous land allocation under uncertain output prices. This study formulates a bi-objective mixed-integer planning model that maximizes expected income and minimizes covariance-based price risk while enforcing empirical crop eligibility, lot occupancy, maturation, and botanical-family recovery. Four publication-complete instances cover one or two farms, 2–8 lots, 5–12 crops, 52–104 planning weeks, and 0.6–6.0 ha. Exact ε-constraint runs in SCIP provide an incumbent reference, standard NSGA-II supplies a low-cost baseline, and two matheuristics apply either full or surrogate-guided continuous quadratic area refinement with SLSQP. Each stochastic method is evaluated over ten seeds using normalized hypervolume, IGD^+, additive epsilon, nondominated-set size, runtime, and refinement effort. Exact-run time rises from 51.25 s on the two-lot instance to 9.52 h on the largest completed instance, while most intermediate SCIP points on the larger instances terminate at the time limit. Surrogate gating reduces refinement calls from 12,000 to 3,000 per replication. On the 8-lot, 104-week instance, it reduces mean total runtime from 1,478.48 to 410.52 s relative to full refinement, but median hypervolume also declines from 0.745 to 0.718. Standard NSGA-II records the highest median hypervolume among the stochastic methods on three of four instances. The evidence therefore supports method selection as a trade-off among reference quality, computation, and planning scale, rather than a universal algorithm ranking.

KEYWORDS
agricultural production planning; multiobjective optimization; NSGA-II; matheuristics; surrogate-assisted optimization; price risk

1. Introduction

Crop planning requires a producer to commit land and timing before harvest prices are observed. Maturation delays, lot capacity, and recovery requirements make these decisions difficult to reverse, while heterogeneous local yields and correlated prices connect agronomic choices to portfolio risk. Production planning must therefore represent not only expected economic return but also the exposure created by the timing and composition of harvests. Reviews of agri-food planning models identify this integration of biological and market constraints as a persistent modelling requirement (Ahumada & Villalobos, 2009). Farm-level diversification studies likewise show why economic risk should be evaluated jointly with return rather than treated as an informal qualification after a plan has been selected (Burbano-Figueroa et al., 2022).

Two computational tensions follow. First, exact multiobjective optimization can generate informative reference trade-offs and solver bounds, but mixed-integer quadratic models become expensive as feasible crop events multiply across lots and weeks. Second, evolutionary methods scale without requiring global mixed-integer solves, but their area decisions may leave exploitable continuous structure unused. NSGA-II remains a standard baseline because nondominated sorting and diversity preservation generate a portfolio of alternatives in one run (Deb et al., 2002). Matheuristics offer an intermediate design: an evolutionary search determines the combinatorial schedule, and mathematical optimization improves continuous decisions conditional on that schedule. Comparable decompositions have been used in multiobjective production scheduling to preserve tractable mathematical structure inside population-based search (Hu et al., 2024).

Continuous refinement itself can dominate runtime when applied to every offspring. Surrogate-assisted evolutionary optimization addresses this allocation problem by learning where expensive calls are likely to be useful, while retaining true objective evaluations for selection and reporting (Liu et al., 2024). This logic is pertinent here because fixing crop identities and harvest weeks leaves a convex quadratic area-allocation problem. A learned gate may reduce the number of conditional refinements, but fewer calls are valuable only if the resulting loss in Pareto quality is acceptable.

This study evaluates that quality–effort trade-off through one reproducible decision model and four solution methods: an exact SCIP ε-constraint procedure, standard NSGA-II, NSGA-II with full continuous refinement, and surrogate-guided H-NSGA-II. The contribution is threefold. The model combines event-based land use with correlated monetary risk and physical activation bounds. The experiment distinguishes certified SCIP points from time-limited incumbents instead of describing the entire exact archive as optimal. Finally, the ablation makes refinement calls and time visible alongside Pareto indicators. The analysis asks: (i) how quickly exact reference computation becomes costly; (ii) how approximate fronts compare with the SCIP incumbent reference; and (iii) whether selective refinement yields a defensible balance between front quality and computational effort.

2. Related Research

2.1. Agricultural production planning under economic risk

Agricultural planning models span tactical production, harvest, distribution, and resource-allocation decisions (Ahumada & Villalobos, 2009). Annual-crop models must additionally coordinate maturation and harvest timing with uncertain yields or markets (Tan & Comden, 2012). A portfolio interpretation is useful because crop choices create joint rather than independent exposure: diversification can reduce risk when price movements are imperfectly correlated. Empirical farm-planning work has operationalized this principle by comparing diversified strategies through return distributions and risk (Burbano-Figueroa et al., 2022). The present model adopts the same economic intuition but embeds it in a weekly event schedule with lot-specific productivity.

The formulation uses price-return covariance rather than independent crop penalties. Monetary exposure depends on both output and harvest-week price, so equal land allocations can create different risk profiles across crops, lots, and weeks. This representation remains a planning approximation: it quantifies correlated output-price

exposure but does not claim to represent every production, weather, or behavioural risk faced by a farm.

2.2. Multiobjective evolutionary optimization

NSGA-II ranks candidates by nondomination and uses crowding distance to retain diversity along the approximated frontier (Deb et al., 2002). Its relevance to agricultural planning is practical: one run returns multiple income–risk alternatives, and decoding can enforce calendar feasibility before objective evaluation. Stochastic performance nevertheless varies across seeds, making replicated quality indicators necessary. Hypervolume summarizes convergence and spread relative to a reference point; IGD^{+} measures the distance from reference points to the approximation; and additive epsilon records the worst shift required for the approximation to weakly dominate the reference.

2.3. Matheuristics and surrogate-assisted optimization

Matheuristics divide a problem so mathematical programming solves a structured subproblem while a heuristic explores the discrete search space. Production-scheduling research demonstrates that mathematical decoding can improve continuous or assignment decisions without repeatedly solving the full mixed-integer model (Hu et al., 2024). In this study, a decoded crop schedule fixes event identities and dates; only cultivated areas remain continuous.

Surrogate assistance adds a resource-allocation layer. Reviews distinguish global, local, hybrid, and learning-based strategies, but a common principle is that predicted utility should determine which expensive evaluations receive budget (Liu et al., 2024). Here an Extra Trees regressor predicts realized refinement gain from candidate features. The surrogate never supplies reported income or risk and never determines archive membership from predicted objectives; it only selects refinement candidates, with a random exploration share retained.

3. Decision Setting and Data

3.1. Study setting and productivity evidence

The decision setting represents Colombian farms divided into heterogeneous lots. A production event assigns a crop, sowing week, harvest week, and cultivated area to one lot. The source productivity table contains 17,059 municipality–crop–planting-month observations, 17 municipalities, and 40 recorded agricultural products. These counts describe the source table; the completed benchmarks use supported subsets of 5–12 crops rather than all 40 products.

Yield estimation combines municipality, crop, and planting-period evidence. Crop eligibility requires at least three supporting observations for a lot's municipality–crop pair; if no crop passes that threshold for a generated lot, the crop with greatest support is retained to avoid a structurally unusable lot. Feasible event variables are created only when the sowing calendar permits the event, harvest occurs within the horizon, the empirical eligibility rule is satisfied, and expected yield is positive.

Table 1. Publication-complete benchmark instances.

Instance	Farms	Lots	Area (ha)	Crops	Weeks
I01_micro	1	2	0.6	5	52
I02_small	1	4	1.8	8	52
I03_diverse	1	8	4.5	12	104
I04_two_farms	2	8	6.0	8	52

3.2. Prices and uncertainty

Weekly prices were constructed for Bucaramanga's Centroabastos market from DANE SIPSA wholesale workbooks for 2018–2026. The input provenance records the downloaded files and hashes. The experiment uses a 104-week price forecast generated from the processed weekly series; shorter instances use its first 52 weeks. A Ledoit–Wolf covariance estimate of crop log returns supplies a positive-semidefinite dependence matrix, with numerical symmetrization and a diagonal adjustment only if required for convexity.

The optimization is performed in Colombian pesos (COP). Publication tables and figures apply a frozen reporting rate of 3,306.86 COP/USD dated 28 September 2026. If p_{kt} is COP/kg, then $p_{kt}^{\mathrm{USD}} = p_{kt}/3306.86$. Likewise, income in million COP is converted to thousand USD by multiplication by 1000/3306.86, and risk based on million-COP exposure is converted by $(1000/3306.86)^2$. Positive constant rescaling leaves Pareto dominance unchanged; the quadratic risk unit changes by the square of the monetary conversion factor.

No `demand_weekly.csv` file was present in the reproducible computational snapshot. Consequently, the reported experiment has no market-throughput cap and makes no use of SIPSA-A demand.

3.3. Benchmark construction

The benchmark ladder changes farms, lots, area, crop count, and horizon under fixed instance seeds. Only instances with all expected files were admitted: one exact run and ten seeds (2026–2035) for each stochastic method. The snapshot scanner reads each `metadata.json`, requires `status=ok`, and verifies the expected front before inclusion. It does not rely on a final run index. At the analysed snapshot, 4 instances meet this rule; the largest contains 8 lots.

4. Mathematical Formulation

The decision problem selects agronomically feasible crop events and their areas. Expected income is maximized while correlated price risk is minimized. The formulation is expressed in the internal monetary units used by optimization; the ex-post USD transformation above is used for manuscript reporting.

4.1. Sets and Parameters

Let $\mathcal{F}$ be the farms, $\mathcal{L}$ the lots, and $\mathcal{L}_f \subseteq \mathcal{L}$ the lots belonging to farm f. Let $\mathcal{K}$ denote crops, $\mathcal{T}$ weekly planning periods, and $\mathcal{T}_k \subseteq \mathcal{T}$ feasible harvest periods for crop k. For an optional market extension, $\mathcal{G}$ denotes sales groups. The sets $\mathcal{C}_{lw}$ and $\mathcal{C}_{lw}^{q}$ contain,

respectively, crop–harvest events that occupy lot l during week w and events from botanical family q whose occupation or recovery intervals conflict in that week.

Parameter A_l is lot area (m^2); n_k is crop maturation duration (weeks); $s(k,t) = t - n_k$ is the sowing week implied by harvest t; r_{lks} is expected yield (kg/m^2); and p_{kt} is expected price (COP/kg). The indicator b_{lk} equals one when empirical evidence makes crop k eligible for lot l. Matrix $\Sigma \succeq 0$ is the covariance matrix of crop price log returns, $q(k)$ is botanical family, and $g(k)$ is sales group. The optional parameter D_{gt} would cap group-g throughput in week t, but is absent from the reported base experiment.

4.2. Decision Variables

For every feasible event (l,k,t), define

$$y_{lkt} = \begin{cases} 1, & \text{if crop } k \text{ is activated on lot } l \text{ for harvest in week } t, \\ 0, & \text{otherwise,} \end{cases} \tag{1}$$

and let $z_{lkt} \geq 0$ be cultivated area (m^2). Aggregate harvested quantity is $x_{kt} \geq 0$ (kg). Monetary exposure is $e_{kt} \geq 0$, measured in million COP, and $\rho \geq 0$ is the epigraph variable used for quadratic risk in the exact formulation.

4.3. Bi-objective Optimization Model

Production and exposure are aggregated through

$$x_{kt} = \sum_{l \in \mathcal{L}} r_{lk,s(k,t)} z_{lkt}, \qquad k \in \mathcal{K},\ t \in \mathcal{T}_k, \tag{2}$$

$$e_{kt} = \frac{p_{kt} x_{kt}}{10^6}, \qquad k \in \mathcal{K},\ t \in \mathcal{T}. \tag{3}$$

The two objectives are

$$\max \quad F_1 = \sum_{k \in \mathcal{K}} \sum_{t \in \mathcal{T}} e_{kt}, \tag{4}$$

$$\min \quad F_2 = \sum_{t \in \mathcal{T}} \mathbf{e}_t^{\mathsf{T}} \Sigma \mathbf{e}_t, \tag{5}$$

where F_1 is expected income in million COP and F_2 is correlated price risk in (million COP)^2. The latter is a monetary mean–variance measure, not a production cost.

Activation uses the physical lot area rather than an arbitrary large constant:

$$0 \leq z_{lkt} \leq A_l y_{lkt}, \qquad l \in \mathcal{L},\ k \in \mathcal{K},\ t \in \mathcal{T}_k. \tag{6}$$

Weekly conflict cliques enforce exclusive lot occupation,

$$\sum_{(k,t) \in \mathcal{C}_{lw}} y_{lkt} \leq 1, \qquad l \in \mathcal{L},\ w \in \mathcal{T}, \tag{7}$$

and family-specific cliques extend this restriction through the configured recovery interval,

$$\sum_{(k,t)\in\mathcal{C}^q_{lw}} y_{lkt} \le 1, \qquad l \in \mathcal{L},\ w \in \mathcal{T},\ q. \tag{8}$$

Variables exist only if $b_{lk} = 1$, $s(k,t)$ is an allowed sowing week, t lies within the horizon, and $r_{lk,s(k,t)} > 0$. This event generation enforces empirical eligibility and temporal feasibility without adding redundant zero variables.
Optional market-throughput extension. If independent market-capacity evidence is available, the implementation can add

$$\sum_{k:g(k)=g} x_{kt} \le D_{gt}, \qquad g \in \mathcal{G},\ t \in \mathcal{T}. \tag{9}$$

Constraint (9) was inactive in every result reported here.

4.4. Epsilon-Constraint Reference Formulation

The reference procedure first solves

$$I^{\max} = \max\{F_1 : (2) - -(8)\}. \tag{10}$$

For each configured $\eta \in \{0, 0.1, \ldots, 1\}$, SCIP then solves

$$\begin{aligned} \min \quad & \rho, \\ \text{s.t.} \quad & \rho \ge F_2, \\ & F_1 \ge \eta I^{\max}, \qquad\qquad (2) - -(8). \end{aligned} \tag{11}$$

When a feasible incumbent is found, an optional strong-Pareto tie break maximizes F_1 subject to the attained risk plus a numerical tolerance. Solver status and relative gap are retained for every point, so a feasible time-limited incumbent is not relabelled as certified optimal.

5. Solution Methods

5.1. Exact epsilon-constraint SCIP

SCIP solves the mixed-integer quadratically constrained formulation with 11 income thresholds. Instance-specific limits are 300, 600, 1,200, and 1,800 s per solve for I01–I04, four threads, and a target relative gap of 10^{-4}. Because the strong-Pareto tie break can require a second solve, total run time can exceed 11 times the point limit.

5.2. NSGA-II

The baseline encodes crop sequences and waiting decisions by lot. Decoding produces feasible sow–harvest events and area values, after which true income and risk are evaluated. Binary tournament selection, crossover probability 0.90, mutation, nondominated

sorting, and crowding-based environmental selection operate with population 80 for 150 generations. Ten fixed seeds permit replication without changing instance data.

5.3. Full continuous quadratic refinement

Conditional on a decoded discrete schedule, area allocation has a linear income expression, a positive-semidefinite quadratic risk expression, physical lot constraints, and fixed event timing. The reported NSGA-II+refinement experiment sends every offspring to this continuous subproblem. Although its mathematical structure is a convex quadratic refinement, the computational backend in this experiment is SLSQP, not SCIP.

5.4. Surrogate-assisted H-NSGA-II

H-NSGA-II uses the same evolutionary operators and SLSQP refinement, but limits refinement to 25% of offspring. During five warm-up generations, candidates are sampled for refinement. After at least 30 realized gains are available, an Extra Trees regressor ranks candidates; 20% of each refinement budget remains random exploration. All selection, nondominated sorting, and reported objectives use actual decoded or SLSQP-refined values. Surrogate predictions only control access to refinement.

6. Computational Experiment

6.1. Experimental settings

All results come from `results/publication_slsqp_test`. Complete instances contain one exact run and 30 stochastic runs: ten each for NSGA-II, full refinement, and H-NSGA-II. Incomplete instances are excluded as units; a finished seed cannot enter tables unless every required seed and method for that instance is present. Optimization outputs remain unchanged, and manuscript currency conversion is reporting-only.

6.2. Performance indicators

For I01–I04, the nondominated SCIP incumbent archive defines the normalization and reference set. This label is deliberate because time-limited points may have nonzero or uninformative gaps. Hypervolume uses reference point $(1.1, 1.1)$ after objective normalization; higher values indicate greater dominated normalized area. Lower IGD^{+} and additive epsilon are preferred. Nondominated point count, total runtime, refinement calls, cumulative refinement time, and useful-refinement hit rate complement these quality measures. Medians summarize stochastic quality and runtime; mean runtime is used only for explicitly stated runtime ratios.

7. Results

7.1. Exact tractability

Total exact-run time increases from 51.25 s for I01 to 9,185.10 s (2.55 h) for I02, 22,824.80 s (6.34 h) for I03, and 34,274.58 s (9.52 h) for I04. Status evidence changes

Table 2. SCIP epsilon-constraint run and point status.

Instance	Runtime (s)	Points	Optimal	Gap limit	Time limit	Max gap
I01_micro	51.25	11	10	1	0	9.85e-06
I02_small	9185.10	11	4	0	7	1.773
I03_diverse	2.28e+04	11	1	0	10	2.376
I04_two_farms	3.43e+04	11	1	0	10	1.394

Table 3. Median Pareto quality and runtime; stochastic methods use ten seeds.

Instance	Method	HV	IGD^+	ϵ^+	ND points	Runtime (s)
I01_micro	SCIP	0.925	0.000	0.000	11.0	51.25
I01_micro	NSGA-II	0.863	0.062	0.089	78.0	48.93
I01_micro	NSGA-II + full	0.862	0.063	0.090	78.0	91.51
I01_micro	H-NSGA-II	0.860	0.063	0.089	78.0	106.98
I02_small	SCIP	1.039	0.000	0.000	11.0	9185.10
I02_small	NSGA-II	1.007	0.032	0.109	79.0	55.51
I02_small	NSGA-II + full	1.009	0.030	0.109	78.5	188.85
I02_small	H-NSGA-II	0.977	0.040	0.138	78.5	144.15
I03_diverse	SCIP	1.047	0.000	0.000	11.0	2.28e+04
I03_diverse	NSGA-II	0.831	0.081	0.306	79.5	72.84
I03_diverse	NSGA-II + full	0.745	0.104	0.400	80.0	1454.71
I03_diverse	H-NSGA-II	0.718	0.121	0.421	79.5	416.35
I04_two_farms	SCIP	1.033	0.000	0.000	11.0	3.43e+04
I04_two_farms	NSGA-II	0.831	0.116	0.210	79.0	61.28
I04_two_farms	NSGA-II + full	0.829	0.112	0.222	79.0	372.73
I04_two_farms	H-NSGA-II	0.793	0.127	0.245	79.0	187.86

with scale. Ten of 11 I01 points are optimal and one terminates at the target gap. I02 has four optimal and seven time-limited points. I03 and I04 each have one optimal income-endpoint and ten time-limited points. Thus, SCIP remains valuable as an incumbent reference, but the experiment does not establish full-front optimality beyond the smallest instance.

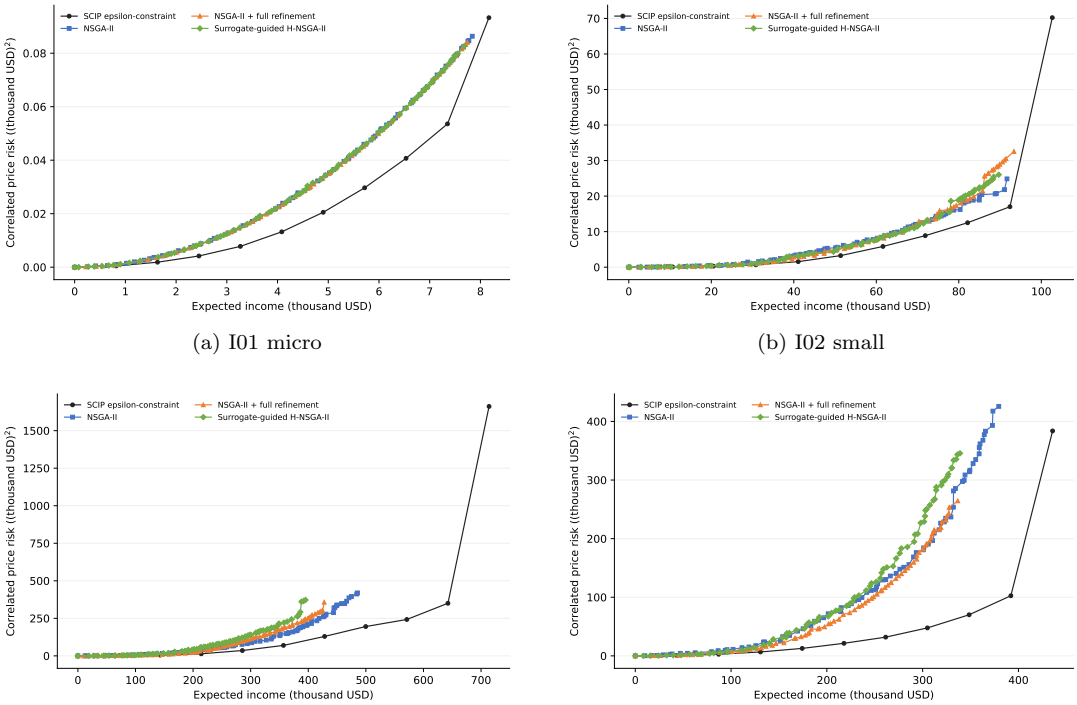
7.2. Pareto quality

Table 3 reports seed medians. Standard NSGA-II has the highest stochastic median hypervolume on I01 (0.863), I03 (0.831), and I04 (0.831). Full refinement is marginally highest on I02 (1.009 versus 1.007 for NSGA-II), with slightly lower median IGD^+ (0.030 versus 0.032). These differences are descriptive, not evidence of a universal ordering. On I03, full and selective refinement attain lower median hypervolume than NSGA-II, while on I04 full refinement is close to NSGA-II and H-NSGA-II is lower. Conditional area optimization therefore does not translate automatically into a better final archive under fixed evolutionary settings.

Figure 1 shows representative seeds chosen nearest each method's median hypervolume. The SCIP curves are incumbent archives; points with time-limit status must not be read as proofs of global efficiency.

7.3. Computational scaling

NSGA-II median runtime stays between 48.93 and 72.84 s across completed instances, whereas exact runtime grows by nearly three orders of magnitude. Full refinement raises median runtime to 1,454.71 s on I03 and 372.73 s on I04. H-NSGA-II reduces those medians to 416.35 and 187.86 s, respectively. Using means, full refinement is 3.60 times slower than H-NSGA-II on I03 and 1.97 times slower on I04.



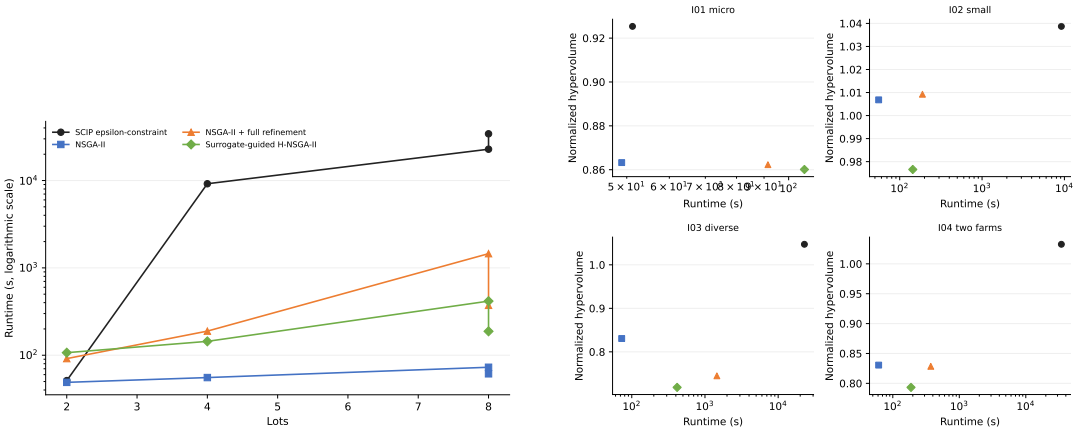
(a) I01 micro

(b) I02 small

(c) I03 diverse

(d) I04 two farms

Figure 1. Income–risk fronts in publication USD units. Stochastic lines show a median-hypervolume representative seed.



(a) Runtime scaling

(b) Quality versus time

Figure 2. Computational effort and normalized hypervolume across complete instances.

Table 4. Full versus surrogate-guided SLSQP refinement.

Instance	Method	Calls	Refinement time (s)	Useful (%)	HV
I01_micro	NSGA-II + full	1.20e+04	29.36	62.3	0.862
I01_micro	H-NSGA-II	3000	9.69	86.2	0.860
I02_small	NSGA-II + full	1.20e+04	114.72	90.4	1.009
I02_small	H-NSGA-II	3000	33.92	98.0	0.977
I03_diverse	NSGA-II + full	1.20e+04	1346.04	96.0	0.745
I03_diverse	H-NSGA-II	3000	284.46	98.8	0.718
I04_two_farms	NSGA-II + full	1.20e+04	289.67	94.8	0.829
I04_two_farms	H-NSGA-II	3000	71.54	98.9	0.793

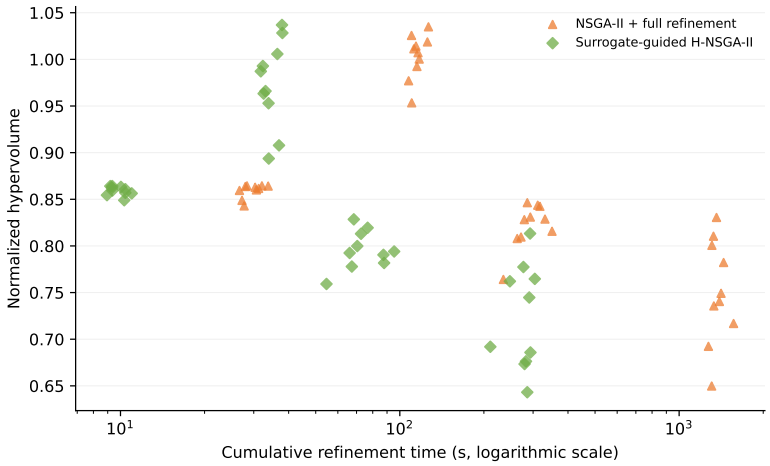


Figure 3. Normalized hypervolume against cumulative SLSQP refinement time for full and surrogate-guided refinement.

7.4. Value of full versus surrogate-guided refinement

Full refinement makes 12,000 SLSQP calls per replication; surrogate gating makes 3,000, a 75% reduction. On I03, median cumulative refinement time falls from 1,346.04 to 284.46 s and useful-refinement hit rate rises from 96.0% to 98.8%. On I04, refinement time falls from 289.67 to 71.54 s and hit rate rises from 94.8% to 98.9%. The higher hit rate shows concentrated allocation, but quality must be read jointly: median hypervolume falls from 0.745 to 0.718 on I03 and from 0.829 to 0.793 on I04. Surrogate gating is therefore a computational economy with a measured quality cost in these instances.

7.5. Interpretation of a representative production schedule

A decision-level example was selected from the median-hypervolume H-NSGA-II seed for I02. Within that front, the compromise minimizes Euclidean distance to the normalized ideal of maximum income and minimum risk. The selected point yields expected income of 59.46 thousand USD and risk of 7.52 (thousand USD)^2. Figure 4 translates that point into 11 production events over four lots. The plan sequences white maize, green pea, and potato; allocated event areas range from approximately 0.03 to 0.70 ha. The diagram demonstrates that a Pareto point is not only an objective pair but an implementable calendar of sowing, harvest, and cultivated-area decisions.

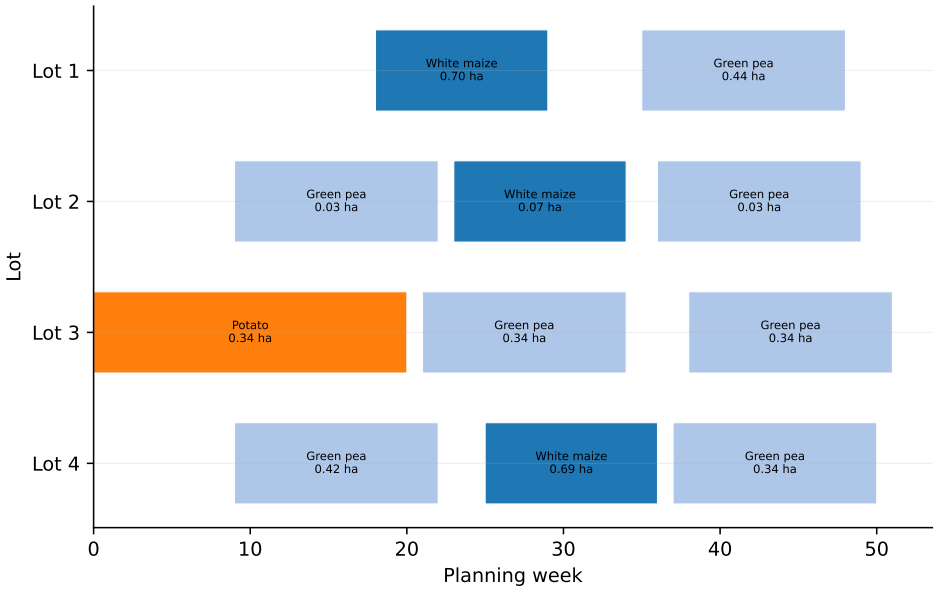


Figure 4. Representative compromise schedule for I02 small. Labels report crop and allocated area; each bar spans sowing to harvest week.

8. Discussion

The experiment separates three computational roles. SCIP supplies economically interpretable reference points and status information, but its rapid runtime growth and the prevalence of time-limited incumbents restrict certification beyond I01. Standard NSGA-II supplies diverse fronts at low and comparatively stable cost. Conditional continuous refinement exploits convex area structure, yet more local optimization does not guarantee a better global archive because refinement changes the search trajectory and consumes time without expanding the discrete schedule space.

Surrogate-guided refinement succeeds at its immediate task: it directs one quarter as many SLSQP calls toward candidates with a higher useful-refinement rate. The larger completed instances show the practical value of that allocation through substantial runtime reductions. However, lower hypervolume and worse distance indicators relative to full refinement show that some omitted calls still matter to the final front. The defensible interpretation is therefore conditional. When runtime is constrained, selective refinement offers a tunable intermediate method; when Pareto quality relative to the incumbent reference dominates computation, this experiment gives no basis for preferring H-NSGA-II over the alternatives.

The comparison also cautions against treating local-refinement success as final-front success. Hit rates exceed 98% for H-NSGA-II on I02–I04, yet its archive can remain worse because a locally useful area change may not improve global spread, and repeated exploitation can alter diversity. Future tests should vary the refinement fraction, exploration share, and gain target, and should compare equal wall-clock budgets rather than only equal generations.

Several limits bound inference. The experiment contains four complete instances and ten stochastic seeds per method. Exact reference archives for I02–I04 contain time-limited points, so quality indicators measure agreement with an incumbent reference, not distance to a proven Pareto set. The covariance objective captures correlated

price exposure but excludes weather, yield, and behavioural uncertainty. The market-throughput extension was inactive because no reproducible demand file was present. Finally, the 17,059-row productivity source has uneven municipality–crop support; empirical eligibility reduces extrapolation but does not remove data sparsity.

9. Conclusion

This study formulates weekly agricultural production planning as a bi-objective event model with discrete crop calendars, continuous cultivated area, empirical crop eligibility, and correlated monetary price risk. A publication-safe experiment on four complete instances compares SCIP ε-constraint computation, NSGA-II, full SLSQP area refinement, and surrogate-guided selective refinement.

The evidence reveals a three-way trade-off. Exact computation provides status-bearing reference points but grows from seconds to hours and loses full-front certification as scale increases. Standard NSGA-II is the lowest-cost stochastic method and achieves the highest median hypervolume among stochastic alternatives on three instances. Surrogate gating cuts refinement calls by 75% and substantially reduces refinement and total runtime on the larger completed instances, but the savings coincide with lower median Pareto quality than full refinement. Method choice should therefore reflect the value assigned to certification, frontier quality, and response time. Larger benchmark conclusions require completion of all prescribed seeds and must be regenerated through the same snapshot rules rather than inferred from partial runs.

Data availability statement

The local reproducibility package contains source code, configuration, input provenance, benchmark metadata, result fronts, seed-level metrics, and the scripts used to regenerate every numerical table and figure. Redistribution of source data remains subject to its original provider terms.

Disclosure statement

No potential conflict of interest is reported by the author.

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